

Team Charter

Team Name	Spokespeople
Active Vision team	Tao Lu Zhiyuan Lu
Project Name	Duration
Active Vision	S2 2024 – S1 2025

Purpose and objectives

This team exists to develop an application that uses advanced computer vision and machine learning techniques to analyze tennis swing dynamics. The expected achievements are listed below:

- Develop Analytical Algorithms: Create robust algorithms to analyze captured motion data, identifying key performance indicators (KPIs) such as swing speed, angle, and impact accuracy.
- Real-Time Feedback: Implement a real-time feedback system to provide immediate insights and suggestions for technique improvement.
- User-Friendly Interface: Develop an intuitive interface for users to interact with the system, view performance data, and track progress.

Roles and responsibilities

1. Tao Lu
 - a. Strengths and skills: Python, Pytorch, Computer Vision, web design, Project management
 - b. Roles and duties: Data pre-processing, algorithm developer, Customer Service Representative
2. Zhiyuan Lu
 - a. Strengths and skills: Python, PyTorch, Tensorflow, OpenCV, C/C++, Linux System
 - b. Roles and duties: Algorithm development, testing
3. Xi Ding
 - a. Strengths and skills: Python, Pytorch, Expertise in Computer Vision and Machine Learning, and Collaborative and communication abilities
 - b. Roles and duties: Data Analyst, algorithm developer0
4. Linxiang Hu
 - a. Strengths and skills: Java, Python, C, HTML, CSS, JS, node.js, Pytorch, Spring and Spring boot Framework, Project management tool such as Maven and Gradle.
 - b. Roles and duties: Data set collection, algorithm development, applications development (if conditions allow) and documentation.

5. Xingchen Zhang
 - a. Strengths and skills: communicate with clients and team members, project management experience, data wrangling and PyTorch
 - b. Roles and duties: Player Recognition part theory analysis and collation, documentation, etc.
6. Yichi Zhang
 - a. Strengths and skills: Python, PyTorch, HTML, CSS, JavaScript, React, Communication Skills, Project Management Experience
 - b. Roles and duties: algorithm development, model deployment and applications development (if possible), also documentation.
7. Pei Ling Lam
 - a. Strengths and skills: Python, resources finding, deep learning, research, Java, Android Studio
 - b. Roles and duties: Algorithm developer for Ball Tracking subgroup
 - c. Quality assurance testing:
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 - d. Skill development plan:
 - New Skills might be required: leadership, project management, planning, effective communication with clients and team members, Flutter, deeper deep learning model for computer vision
 - Learning goal:
 - Resources:
 - Timeline: dedicated 3 hours per week to enhance professional skills along with the tutorials and workshops; learn Flutter during semester break for application development in the next semester; explore more latest deep learning research papers for computer vision and try out their models.

Skill Development Plan

New skills acquisition

Leadership: It will be developed for everyone in the team since we will change the leader weekly so that everyone can boost the leadership.

Project Management abilities: Regular discussions about the structure of our project and tasks will be organized to improve project planning, execution, and monitoring skills, ensuring efficient use of resources and timely completion of project milestones. Besides, everyone will participate in using the management plan tool (Jira) to organize our tasks.

Advanced skills on ball tracking models and relevant machine learning algorithms: Peiling Lam, Tao Lu, Xi Ding (Ball tracking team)

Advanced skills on players tracking models and relevant machine learning algorithms: Yichi Zhang, Lingxiang Hu, Xingchen Zhang, Zhiyuan Lu (Player tracking team)

Quality assurance

Testing Protocols: Testing sessions after the weekly member meeting will be conducted to ensure the newly acquired skills are effectively applied in the project. This includes peer reviews, progress tracking, and feedback sessions to evaluate the integration and impact of new skills.

Evaluation metrics

Peer review: we will review each other's application of new skills in project tasks.

Contribution to the Project: Assess the contribution of acquired skills to project improvements.

Goal of the team

Everyone can enhance their ability to apply computer vision knowledge in practical settings.

We can improve our interpersonal skills and strengthen our ability to cope with setbacks and failures.

Values

We are a team; we share problems and most importantly we share successes.

We are a team; we respect each other by pointing out mistakes/doubts politely.

We take responsibility for our actions as a team.

We can make mistakes without blame, provided lessons are learned and admitted.

We accept disagreements, but we support the decisions completely once they are made.

Norms

Decision-making process: speak out issue and follow the decision log template

Resolving conflict strategies: address conflict, identify the issues that cause conflict, bring the involved parties together to have communication, identify a solution to it.

Feedback providing strategies: though 3 audits will be occurring in each semester, a weekly regular check-in will be done within the team to figure out what works and what not. The feedback will be done on the templates, meeting structure, leading styles, and coding styles.

Communication plan

The team will have weekly client meetings (Wednesday) and member meetings (Friday). Meeting minutes should be made after every meeting session and shared on SharePoint.

Main communication channel: Discord

Informal communication: WeChat group

Files sharing and landing page channel: SharePoint (Quick links to other channels are available [here](#))

Resources and support

Technical support: TIME Lab members, Google

Hardware support (GPU): ANU Lab

General support: Elena Williams (tutor)

Team's Dated Signatures

Team Member	Date
Xi Ding	04/08/2024
Team Member	Date
Xingchen Zhang	04/08/2024
Team Member	Date
Tao Lu	04/08/2024
Team Member	Date
Yichi Zhang	04/08/2024
Team Member	Date
Linxiang Hu	04/08/2024
Team Member	Date
Zhiyuan Lu	05/08/2024
Team Member	Date
Pei Ling Lam	04/08/2024